

Novelty is Socially Constructed: A Multi-Agent Simulation of Scientific Community Dynamics

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Abstract

A central open problem in AI Scientist systems is whether novelty evaluation can be automated. We argue that novelty is not an intrinsic property of a hypothesis but an emergent outcome of community arbitration — the distributed process by which a scientific community positions a claim relative to collective knowledge. To test this, we construct a multi-agent simulation using MiroFish in which heterogeneous reviewer agents with fragmented local memories evaluate AI-generated hypotheses. We show that (1) isolated semantic classifiers systematically misidentify novelty when knowledge is distributed across agents, (2) community consensus accuracy improves non-linearly with knowledge-sharing topology, and (3) human oracle intervention remains necessary above a noise threshold even with strong local classifiers. These results provide direct empirical support for the non-automatability claim in ? and constrain the design space of the V_{human} layer in AI Scientist systems.

1 Introduction

The SCAGENT framework (S, G, E, V, M, π) (?) identifies the completeness of the validator V as the single most consequential design variable in AI Scientist systems. Among the four validator layers — V_{syntax} , V_{semantic} , $V_{\text{empirical}}$, V_{human} — the last is the most resistant to automation. Prior work has argued informally that novelty evaluation requires human judgment (?); we provide the first controlled simulation evidence for this claim.

2	Background
2.1	Novelty as Community Arbitration
2.2	MiroFish: Multi-Agent Social Simulation
3	Simulation Design
3.1	Agents
3.2	Environment
3.3	Experimental Conditions
3.4	Metrics
4	Results
4.1	Isolated Agents Fail at Global Novelty
4.2	Topology Drives Consensus Quality
4.3	Human Oracle Threshold
5	Discussion
6	Conclusion

AI Involvement Checklist

- **Hypothesis development.** The research question (novelty as emergent community property, not intrinsic hypothesis property) was proposed by the human author based on framework analysis in ?. Claude Code helped identify relevant Science of Science literature and formalized the agent decision rules.
- **Experimental design.** The human author designed the five experimental conditions and the latent concept graph environment. Claude Code implemented the simulation codebase (`src/`) and suggested the inter-agent trust topology variants.
- **Analysis.** Claude Code ran parameter sweeps and generated result figures. The human author interpreted results and identified the non-linear threshold behavior.
- **Writing.** Claude Code drafted manuscript sections; the human author wrote the core argument and conclusions.

Responsibility & Reproducibility Checklist

- **Claims and evidence.** All quantitative claims are supported by simulation results reported in §???. Simulation code is released at `src/` in the supplementary material.
- **Limitations.** The hypothesis space is abstract (graph-structured, not drawn from a real scientific domain). Agent personalities are synthetic. Results may not transfer directly to real scientific communities.
- **Reproducibility.** Full simulation code, random seeds, and parameter configurations are included in supplementary material. All experiments can be reproduced on a laptop CPU in under 30 minutes.
- **Data and code availability.** Simulation code released with submission. No external datasets used.
- **Ethical and societal impact.** This work informs the design of AI Scientist systems. The finding that V_{human} cannot be fully automated supports calls for maintained human oversight in autonomous research pipelines.